

Python API Backend Curriculum

Part of the [training document series](#). This folder is a self-contained curriculum for building a Python web API that serves computed analysis results as JSON — the "decoupled" presentation-layer option described in the [architecture overview](#).

Reading order

1. **HTTP, REST & FastAPI Fundamentals** — HTTP methods and status codes, REST conventions, and the FastAPI framework's routing, validation, and dependency-injection model.
2. **API Design & Best Practices** — data loading and caching, error-handling patterns (failing closed on pipeline output), resource/URL design, versioning, pagination, response envelopes, reporting the scope actually served, security fundamentals.
3. **Testing** — `pytest`, FastAPI's `TestClient`, fixture-based test data, mocking, fixtures built by running the real upstream pipeline, and how to verify a test actually catches the regression it claims to.
4. **Deployment** — running the API in production: ASGI servers, containers, cloud platforms, reverse proxies and edge policies, refresh-driven restarts, CI/CD, and health checks.

Prerequisites

Comfort with Python functions, classes, and type hints is assumed. No prior web-framework experience is assumed.

See also

- [Architecture Overview](#) for how this layer fits between the data-processing layer and a frontend
- [Data Science / ML](#) for what typically produces the data this API serves
- [React Front End](#) for a typical human-facing consumer of an API built this way
- [MCP Server](#) for wrapping this same API as tools an LLM agent can call, instead of (or alongside) a frontend